Ashutosh Naik

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EDUCATION

University of Colorado Boulder Boulder, CO
Master of Science in Computer Science GPA: 4.00/4.00 *Expected May 2024*
Relevant Coursework: Datacentre Scale Computing, Design and Analysis of Algorithms, Advanced Robotics

Sardar Patel Institute of Technology Mumbai, India
Bachelor of Technology in Computer Engineering GPA: 9.77/10 *May 2022*
Relevant Coursework: Object Oriented Programming, Artificial Intelligence and Soft Computing,
Distributed Systems, Design and Analysis of Algorithms, Database Management Systems, Advanced Data Structures

TECHNICAL SKILLS

Programming Languages: Python | C++ | C | Go | SQL
Libraries: Docker | Kubernetes | Google Cloud Platform | ROS | PyTorch | TensorFlow | NumPy | OpenCV | MPI

EXPERIENCE

Skinzy Software Solutions, *Machine Learning Intern* *Dec. 2020 – Dec. 2021*
- Achieved an accuracy of over 90% across 15 classes by fine-tuning MRCNN, UNET and detectron2 on custom datasets for segmenting skin diseases.
- Attained an accuracy of over 80% on a jaundice detection task using PyTorch.
- Conducted a session and was in-charge of hiring and training new interns.

PROJECTS

CloudBoard – A Cloud Clipboard
Technologies: Go, Google Cloud Platform APIs, Redis, Docker, SQL. *June 2023*
- Devised a web application to synchronize users' clipboards across devices.
- Coded a JWT authentication system for user authentication and utilized Websockets for clipboard data transfer.
- Developed REST APIs with Gin and deployed it on Google Cloud Run.

TowMater – Autonomous 1/10th Scale Car
Technologies: C++, ROS, OpenCV, Robotics. *Jan. 2023 – May 2023*
- Collaborated on an autonomous car project in a team of 6.
- Integrated Intel Realsense 2 camera and IMU to gather depth data and maintain balance and stability.
- Formulated a PID controller for precise vehicle control and added ability to take jumps under 1 meter wide.

ShelfHelp
Technologies: Python, PyTorch, Unsupervised Learning, Siamese Networks. *Aug. 2022 - April 2023*
- Created a system to assist visually impaired with grocery shopping using computer vision and guidance algorithms.
- Implemented metric collection framework and analysed data, for key observations.
- Utilized CRAFT model for text detection and recognition and worked with SimCLR and BYOL models for image representation learning.

Image Augmentation as a Service
Technologies: Python, Flask, Kubernetes, Docker, Google Cloud Platform APIs. *Nov. 2022*
- Designed an application to apply image transformations as a service.
- Implemented augmentation functions using Google Cloud Functions and stored images in Google Cloud Storage.
- Developed REST APIs with Flask and deployed it on a Kubernetes Cluster.

Segmentation of Nuclei
Technologies: Python, OpenCV, PyTorch. *Jan. 2022 – April 2022*
- Pre-processed nuclei segmentation datasets obtained using OpenCV, NumPy and scikit-image.
- Fine-tuned the TransUnet model using PyTorch on the datasets to achieve a 64% higher IOU score than the U-Net model.

PUBLICATIONS/CERTIFICATIONS

- "ShelfHelp: Empowering Humans to Perform Vision-Independent Manipulation Tasks with a Socially Assistive Robotic Cane" Published at AAMAS 2023.